# A First Lean Formalization Audit of the IUT III Corollary 3.12 Corridor

IUT Lean formalization project

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**Abstract**

This note records the present state of a Lean formalization of a narrow part of Mochizuki's inter-universal Teichmuller theory: the passage from IUT III, Theorem 3.11 to Corollary 3.12, with attention to the Scholze–Stix objection. The current Lean code proves conditional route theorems, obstruction lemmas, and the final ordered-real algebra in Step (xi-f). It does not prove Corollary 3.12 and does not refute it. The main formal result so far is that a supplied hull/log-volume $q$-to-$\Theta$ comparison gives $C_\Theta \geq -1$, and that a nonarchimedean (Ind3) entry together with explicit Hodge/SHE square-weight transport data, or the primitive factored square/full-label SHE obligations, feeds such a comparison, while several simplified collapse mechanisms are rejected by Lean.

## 1 Scope

The review used the local Markdown conversions of IUT I–IV, Mochizuki's 2026 formalization note, and the Scholze–Stix note. The relevant source passages are:

- IUT I: initial $\Theta$-data, prime strips, Hodge theaters, and non-scheme-theoretic $\Theta$-links.
- IUT II: Hodge–Arakelov evaluation, theta values of shape $q^{j^2}$, multiradiality, and Gaussian monoids.
- IUT III: Theorem 3.11; Corollary 3.12, especially Steps (x) and (xi); Remarks 3.12.2 and 3.12.3.
- IUT IV: later explicit log-volume estimates and Diophantine consequences.
- Scholze–Stix: the ordered real-line identification problem in their Section 2.2.
- Mochizuki 2026: the "3.11.5 ⇒ 3.12" fourth-triangle decomposition.

The namespace `Iut.Stage1` is local terminology for the present corridor layer. It is inspired by Mochizuki's staged formalization plan, but does not assert completion of his Stage 1: many Frobenioid, Hodge-theater, log-Kummer, and theta-link constructions remain external obligations.

## 2 Mathematical Target

IUT III Step (x) treats procession-normalized log-volumes as invariants for (Ind1) and (Ind2), and turns (Ind3) into an upper inequality. Step (xi) then uses the multiradial construction, IPL, SHE, holomorphic hulls, determinants, and log-volume to place the $q$-pilot log-volume in an upper ray controlled by the $\Theta$-pilot log-volume:

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|} \quad \Rightarrow \quad -|\log(q)| \leq -|\log(\Theta)|.$$

If $C_\Theta$ is any real number satisfying

$$-|\log(\Theta)| \leq C_\Theta |\log(q)|, \qquad |\log(q)| > 0,$$

then Step (xi-f) concludes $C_\Theta \geq -1$. This last inference is now a Lean theorem. Lean also proves the strict variant: if the q-pilot lies strictly inside the $\Theta$-upper ray, then $C_\Theta > -1$; conversely, the boundary case $C_\Theta = -1$ forces equality of the q-pilot and $\Theta$-boundary log-volumes and places the $\Theta$-boundary on the negative branch. Mochizuki's 2026 note isolates the final comparison as:

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The present formalization targets only this corridor, not the full construction of Theorem 3.11.

## 3 Dispute Interface

Scholze–Stix require explicit bookkeeping for all ordered one-dimensional real vector spaces. Their pressure point is that the $\Theta$-side concrete objects carry $j^2$-scaled degrees, while a consistent identification of the real lines appears to leave no room for inserting all these scalars.

The formalization therefore separates:

$$\begin{array}{c} \text{representative } j^2 \text{ data} \\ \text{balanced sign-compatible data} \\ \text{aggregate averaged data} \\ \text{final hull/log-volume data} \end{array} \quad \text{and} \quad \begin{array}{c} \text{raw real equality} \\ \text{transported equality} \\ \text{matched transport scales} \\ \text{cancellable ordered-line comparison.} \end{array}$$

This separation is the main correctness guard in the current code.

## 4 Lean Coverage

The relevant implementation is in `Iut/Stage1/IUTStage1SourceCore.lean`, `Iut/Stage1/IUTStage1Remark312Absorption.lean`, `Iut/Stage1/IUTStage1Source.lean`, and `Iut/Stage1/IUTStage1Experiments.lean`. The code currently covers:

- labeled real-line copies and positive-scale transports;
- a finite $\mathbb{F}_\ell$-label model with representative square weights $j \mapsto j^2$;
- a representative pilot-degree profile $\deg(\Theta_j) = j^2 \deg(q)$;
- a finite $q^{j^2}$ endpoint connecting this representative Gaussian-degree rule to the square-weight profile: for canonical representatives, the Gaussian degree is the square weight times the environment/q-pilot degree; Lean packages this together with the zero branch, the canonical $j = 1$ branch, and sign invariance;

- a splitting-monoid/tensor-packet endpoint for the same finite branch: the core generator is zero, each procession generator is the procession square times the environment degree, acting on the tensor packet adds the generator log-volumes, and nonnegative environment degree gives monotonicity of the normalized acted packet; Lean now also identifies the averaged splitting-generator contribution with the full-label Gaussian average, so the normalized acted packet is the original packet plus either of these equivalent finite $q^{j^2}$ averages; normalized-log-volume triviality of the action is equivalent in this finite model to zero environment degree, with strict increase for positive environment degree and strict decrease for negative environment degree;

- proof that one label-independent scale cannot absorb all representative $j^2$ scales in this model;

- proof that, for nonzero $\deg(q)$, one label-independent scale cannot account for all representative $\Theta_j$-degrees; Lean now proves the exact converse classifier: such a scale exists if and only if $\deg(q) = 0$; under the Corollary 3.12 sign hypothesis $0 < -\deg(q)$, no such scalar exists;

- proof that the raw representative $j^2$ theta-degree rule does not descend to the sign quotient $|F_\ell| = F_\ell/\{\pm 1\}$ when $\deg(q) \neq 0$;

- an absolute-label exponent on $|F_\ell| = \{0\} \cup F_\ell^\times/\{\pm 1\}$ which agrees with $j^2$ for the canonical representatives $0 \leq j \leq (\ell - 1)/2$, and the corresponding theta-degree profile; Lean proves that the coordinate map $F_\ell \to |F_\ell|$ is surjective, that the zero absolute label is reached exactly by $j = 0$, hence has singleton coordinate fiber, and that every nonzero sign-label class has a nonzero coordinate representative; Lean also proves the exact fiber relation: two coordinates have the same full absolute label if and only if they are equal or negatives of one another; equivalently, every nonzero full-label fiber is exactly a sign pair $\{j, -j\}$, hence has cardinality 2. These zero, canonical-one, equality-up-to-sign, and surjectivity facts are now packaged as a full-label endpoint. Lean also proves that the half-range procession map to $|F_\ell|$ is bijective, using the minimal absolute representative $j_{\min}$ for surjectivity and the exponent $j^2$ for injectivity; this gives $\#|F_\ell| = (\ell+1)/2$ in the finite model and permits Lean to reindex absolute-label averages by half-range processions; Lean also proves that the half-range procession top is exactly $(\ell - 1)/2$ and that the resulting finite index set has cardinality $(\ell + 1)/2$, matching the $j \in \{0, \ldots, (\ell - 1)/2\}$ convention in IUT II; on this half-range procession, Lean proves the Gaussian degree formula $G(j) = j^2 \deg(q)$, with $G(0) = 0$, and the normalized average identity $6\ \mathrm{Avg}(G) = j_{\max}(2j_{\max} + 1) \deg(q)$; by reindexing, the same identity holds for the average over $|F_\ell|$. Since $j_{\max} \geq 2$, Lean proves that this full average is not the singleton $j = 1$ identity value when $\deg(q) \neq 0$; in the nonpositive environment branch, the full $|F_\ell|$-average is bounded above by $\deg(q)$, and hence by any bound above $\deg(q)$; if $\deg(q) < 0$, this bound is strict: the full average is $< \deg(q)$, while for $\deg(q) > 0$ it is $> \deg(q)$; the same facts are now packaged as a typed $|F_\ell|$-indexed procession-normalized average object, where the average is provably distinct from the canonical $j = 1$ component unless $\deg(q) = 0$, strictly below it if $\deg(q) < 0$, strictly above it if $\deg(q) > 0$, and equal to that component if and only if $\deg(q) = 0$;

- the degree-level Gaussian evaluation $q \mapsto (q^{j^2})_{j \in |F_\ell|}$, including identity at $j = 1$; Lean proves that the absolute-label exponents at $j = 1$ and $j = 2$ are 1 and 4, hence the corresponding theta/Gaussian degrees are distinct whenever $\deg(q) \neq 0$, and equal exactly in the degenerate case $\deg(q) = 0$; Lean now also converts such an evaluation into a cusp/full-label log-volume compatibility object and proves the full-label value-preservation branch when the source and target environment degrees agree. Conversely, Lean proves that full-label value preservation for two Gaussian evaluations is equivalent to equality of their environment degrees, and this equality is already detected at the canonical $j = 1$ component. Lean also proves the IUT II restriction check at the canonical $j = 1$ sign label: the Gaussian component and the corresponding cusp-class log-volume are both the original environment degree; Lean also records the accompanying IUT II $F_\ell$-label model endpoint: the carrier has cardinality $\ell$, the maps to and from $\mathbb{Z}/\ell\mathbb{Z}$ are inverse, and the additive torsor action is transported by coordinate addition. Lean also packages the unit/sign bridge: the distinguished sign unit acts by negation, the $\{\pm 1\}$-unit subgroup orbits are exactly sign orbits, and the canonical cusp class is the sign class of coordinate 1. It also records the zero/nonzero local-object assignment and the coordinate-to-full-label log-volume compatibility endpoint, including invariance under $j \mapsto -j$; Lean now packages the Remark 3.12.2(i)(b) finite-symmetry split in one endpoint: additive $F_\ell$-translation, multiplicative unit action, the sign-unit orbit criterion, equality up to sign in the full-label quotient, and compatibility of unit action with coordinate-to-full-label descent, and the warning in the finite model: the singleton restriction to $j = 1$ is not stable under the additive $F_\ell$-torsor action, since translation by 1 sends 1 to $2 \neq 1$; more generally, Lean proves that every nonempty translation-stable finite subset of $F_\ell$ is all of $F_\ell$, packaged as a translation-closure endpoint, so no nonempty proper subset restriction is compatible with this symmetry; Lean also proves that no nonzero additive translation descends to a well-defined map on $|F_\ell|$, since the equal absolute labels of $t$ and $-t$ would have to map simultaneously to the nonzero label of $2t$ and to 0; equivalently, an additive translation descends to $|F_\ell|$ if and only if it is the zero translation; the same iff is exposed at the route interface as the exact condition for a translation equivalence to preserve full labels;

- a canonical representative square-weight profile on $F_\ell$: Lean proves the total $j^2$-mass is positive from the $j = 1$ contribution and therefore constructs the profile without an external positivity premise; it also proves $6 \sum_{j \in \mathbb{Z}/\ell\mathbb{Z}} j_{\mathrm{rep}}^2 = (\ell - 1)\ell(2(\ell - 1) + 1)$, hence the corresponding average formula after division by $\ell$;

- the square-weighted constant-average check: a constant procession-normalized log-volume remains equal to the same constant after averaging with the canonical $j^2$-weights;

- the same constant-average check inside the $F_\ell = \mathbb{Z}/\ell\mathbb{Z}$ cusp-label corridor: if the full-label log-volume is constant, the square-weighted average used in the $\Theta$-route is that constant, including for the canonical weights;

- the constant-target structured-SHE subcase: the formal route now proves equality of the square-weighted average with the $\Theta$-source average, not merely an upper bound;

- a log-volume shadow of the IUT III splitting-monoid action: the generator attached to a procession label contributes $j^2 \deg(q)$, contributes 0 at the core label, and adds its finite generator sum to the tensor-packet log-volume; Lean also proves the procession-normalized form: the acted tensor-packet average is the original tensor-packet average plus the averaged Gaussian generator contribution, equivalently the average of the $j^2 \deg(q)$ contributions over the procession container; for $S_{j+1} = \{0, \ldots, j\}$, Lean expands the total generator contribution to $j(j+1)(2j+1)\deg(q)/6$, and the resulting normalized log-volume change to $j(2j+1)\deg(q)/6$; at the final procession length this averaged generator contribution is equal to the full-label Gaussian average; since $j = \lfloor \ell/2 \rfloor \geq 2$, a nonnegative environment degree makes the acted normalized tensor-packet log-volume an upper bound for the original one; equality of acted and original normalized log-volume is equivalent to zero environment degree, and the sign of the environment degree determines the strict direction of the change;

- the finite procession containers $S^{\pm}_{j+1} = \{0, \ldots, j\}$, their nested inclusions, core-label stability, stage cardinality $j+1$, and total procession ambiguity $(\ell^{\pm})!$ rather than $(\ell^{\pm})^{\ell^{\pm}}$;

- the map from the final procession container to $|F_\ell|$, sending the core label to 0 and recovering the exponent $j^2$;

- the log-volume shadow of the tensor packet over $S^{\pm}_{j+1}$, with each factor a finite-fiber direct sum over places above $v_{\mathbb{Q}}$, total log-volume given by the finite sum over the container, normalized denominator $j+1$, and invariance under container permutations;

- source-side local log-volume interfaces: Lean packages finite local log-volume identification, positive capsule count, procession-normalized averaging, and local container estimates $q_{\text{signed}} \leq \text{logvol}_{\text{local}}$, while leaving the analytic construction of the local objects external;

- the $F_\ell$-label average specialization: for labels represented as $\mathbb{Z}/\ell\mathbb{Z}$, Lean rewrites the procession-normalized average denominator as $\ell$;

- the zero/nonzero Gaussian average split: Lean proves

$$\#(\mathbb{Z}/\ell\mathbb{Z})^\times = \ell - 1, \qquad \operatorname{Avg}_{j \in F_\ell} G(j) = \frac{\ell - 1}{\ell} \operatorname{Avg}_{j \neq 0} G(j),$$

  for the degree-level Gaussian evaluation $G$, using $G(0) = 0$. Thus the paper's uniform $1/\ell$ weighted-volume convention is not silently replaced by the auxiliary nonzero-only average; in the negative signed branch Lean also proves the strict inequality $\operatorname{Avg}_{j \neq 0} G(j) < \operatorname{Avg}_{j \in F_\ell} G(j)$, while in the positive branch the inequality is reversed; equality between the two averages holds if and only if $\deg(q) = 0$. Lean also proves that the all-$F_\ell$ coordinate average itself is zero if and only if $\deg(q) = 0$; Lean separately formalizes the nonzero half-range $j = 1, \ldots, (\ell-1)/2$ average, proves that the signed nonzero carrier $F_\ell^\times$ is the twofold cover of this half-range by $j$ and $-j$, and therefore proves that the signed nonzero average equals the nonzero absolute-label average. It also proves $6 \operatorname{Avg}_{j \neq 0, |F_\ell|} G(j) = (j_{\max} + 1)(2j_{\max} + 1) \deg(q)$; Lean also proves the exact rescaling

$$\operatorname{Avg}_{|F_\ell|} G = \frac{j_{\max}}{j_{\max} + 1} \operatorname{Avg}_{j \neq 0, |F_\ell|} G.$$

  Since the full $|F_\ell|$-average has coefficient $j_{\max}(2j_{\max} + 1)/6$, Lean proves that the nonzero absolute-label average is strictly below the full average when $\deg(q) < 0$ and strictly above it when $\deg(q) > 0$, with equality if and only if $\deg(q) = 0$. Combining the signed and absolute computations, Lean proves

$$\operatorname{Avg}_{F_\ell} G = \frac{j_{\max}(j_{\max} + 1)}{3} \deg(q),$$

  equivalently $\operatorname{Avg}_{F_\ell} G = ((\ell+1)/\ell) \operatorname{Avg}_{|F_\ell|} G$, so $\operatorname{Avg}_{F_\ell} G$ is strictly below $\operatorname{Avg}_{|F_\ell|} G$ when $\deg(q) < 0$, strictly above it when $\deg(q) > 0$, and equal to it only when $\deg(q) = 0$;

- the IUT II Remark 4.7.3 zero/nonzero weighted-volume relation: Lean records a relation in which every nonzero full label is subordinate to the zero label, proves that $0 \ll 0$ is false, proves that this relation is not symmetric at the zero label, and proves the exact criterion $a \ll 0 \iff a \neq 0$ for full labels; at coordinate level this becomes $j \ll 0 \iff j \neq 0$, so the all-coordinate statement is false precisely because it includes $j = 0$. Lean packages this with unit-action preservation and the zero-translation criterion for affine preservation. Lean also proves that the full-label set subordinate to zero has cardinality $(\ell-1)/2$, the nonzero absolute half-range, and decomposes any full-label sum, hence any full-label average, into its zero contribution plus the subordinate nonzero contribution with denominator $(\ell+1)/2$. Applying this to the Gaussian degree function, whose zero component is 0, Lean rewrites the full absolute-label Gaussian average as the subordinate nonzero Gaussian sum divided by the full absolute-label denominator and proves the closed form

$$6\sum_{a\ll 0} G(a) = j_{\max}(j_{\max}+1)(2j_{\max}+1)\deg(q).$$

Lean also proves that this subordinate sum is exactly the canonical nonzero absolute half-range sum $\sum_{j=1}^{j_{\max}} G(j)$. In contrast to nonzero additive translations, Lean defines the multiplicative unit action on full labels, proves its compatibility with the coordinate formula $j \mapsto a \cdot j$, and proves that it preserves the subordinate zero/nonzero relation. Lean also proves the unit and multiplication laws for this action and proves that the sign subgroup $\{\pm 1\}$ acts trivially on full absolute labels. Each unit action is packaged as an equivalence of full labels with inverse action by $a^{-1}$, and Lean proves that the zero label is fixed exactly. Consequently Lean proves that full-label sums and full-label averages are invariant under this multiplicative permutation; the same holds for sums over the subordinate nonzero full-label subset. These invariance statements are also specialized to the Gaussian degree function $G(a)$, so the full Gaussian average and the subordinate Gaussian sum are invariant under multiplicative reindexing. At the coordinate level, Lean proves the corresponding sharp contrast: unit multiplication preserves the predicate $j \ll 0$, whereas additive translation preserves this predicate for all $j$ if and only if the translation is zero. More strongly, an affine coordinate operation $j \mapsto aj+t$, with $a \in F_\ell^\times$, descends to full absolute labels if and only if $t=0$, and preserves the subordinate predicate for all $j$ if and only if $t=0$. Lean now packages the unit-action laws, coordinate compatibility, zero-label preservation, and affine descent criterion in one endpoint. Lean separately proves that raw $F_\ell$-coordinate sums are invariant under additive translation, unit multiplication, and their affine composition; this is only finite-permutation invariance and does not supply descent to $|F_\ell|$. For the Gaussian coordinate average, Lean packages this as a diagnostic: every affine map with nonzero additive offset preserves the raw coordinate average but fails full-label descent. If the Gaussian environment degree is nonzero, the same affine operation also changes the zero Gaussian component, so aggregate average invariance is not pointwise Gaussian compatibility. Lean also proves the exact degeneracy criterion: a coordinate Gaussian component is zero if and only if its coordinate is zero or the environment degree is zero; for an affine zero-coordinate image this becomes $t=0$ or $\deg(q)=0$. At the purely labelwise level, Lean proves the exact pointwise preservation criterion: $j \mapsto aj+t$ preserves the full absolute label of every coordinate if and only if $t=0$ and $a \in \{\pm 1\}$. At the Gaussian-value level, Lean proves the exact nondegenerate pointwise criterion: if the environment degree is nonzero, then $j \mapsto aj+t$ preserves all coordinate Gaussian degrees if and only if $t=0$ and $a \in \{\pm 1\}$. Equivalently, in this finite affine family, nondegenerate pointwise Gaussian preservation is exactly preservation of the full absolute-label map. Lean also proves the degenerate complement: if $\deg(q)=0$, then every Gaussian degree is 0, so every affine coordinate operation preserves Gaussian values. Combining both cases, Lean proves the exact classification: affine Gaussian-value preservation is equivalent to either $\deg(q)=0$ or preservation of the full absolute-label map. Uniform preservation over all nonzero environment degrees is therefore equivalent exactly to full absolute-label preservation. For the stricter representative $j^2$ square-weight branch, Lean proves the corresponding affine criterion $t = 0$ and $a = 1$. Thus the sign action is legitimate for the balanced/full-label Gaussian branch, but not for the representative-square transport branch. Requiring both the representative-square branch and the full-label branch is therefore equivalent, in the unit-affine family, to the identity case $t = 0, a = 1$. Likewise, nondegenerate pointwise Gaussian preservation plus representative square preservation is equivalent to $t = 0, a = 1$. Lean exposes this as a direct diagnostic: the affine action $j \mapsto -j$ preserves Gaussian degrees pointwise but does not preserve the representative $j^2$ coordinate-square profile. Lean now packages the sign-negation case as a source-layer transport theorem: the balanced self-transport $j \mapsto -j$ simultaneously preserves all Gaussian degrees and all balanced square weights, while it fails preservation of the representative constant-one square-weighted summands. This is a formal separation between sign-compatible balanced transport and the stricter representative branch. A further experiment theorem combines this preservation/failure package with the final-route audit: the same balanced negation transport is not the comparison level consumed by the final weighted-theta route. Lean also converts the zero/nonzero cusp-label compatibility object to a uniform $F_\ell$ average and proves that a diagonal constant log-volume distribution averages to that same constant; the same compatibility layer now exposes, as a single endpoint, the equality between coordinate-normalized log-volumes and full-label log-volumes, together with the zero branch, nonzero cusp-class branch, and sign-pair invariance; Lean also proves that the uniform $F_\ell$-average is the average of these full-label values, and that zero/cusp-class upper bounds propagate to this average;

- the IUT III Step (x) averaged-log-volume corridor: Lean records that (Ind1) and (Ind2) preserve the procession-normalized averaged log-volume, while (Ind3) supplies an upper bound; the code now constructs this corridor from three $F_\ell$-cusp log-volume compatibility objects when the after-(Ind2) zero and cusp-class branches are bounded by the proposed (Ind3) bound; the before-, after-(Ind1)-, and after-(Ind2)-averages are all bounded by the same (Ind3) upper bound; Lean now packages the Remark 3.12.2(iii) log-link-juggling endpoint by combining these two equalities with the one-sided (Ind3) bound, and separately records the matching nonarchimedean/archimedean place families for the upper-semi state;

- the Step (x) to Step (xi) upper-ray handoff: if the q-pilot reading is the pre-indeterminacy averaged log-volume and the $\Theta$-hull reading is the (Ind3) upper bound, Lean constructs the hull/determinant upper-ray datum used in Step (xi); the code now also constructs this handoff directly from the $F_\ell$-cusp compatibility corridor plus the determinant/hull identifications and proves the resulting q-pilot membership in the hull upper ray; Lean now exposes the defining equivalence $x \in R_{\leq \Theta} \iff x \leq \Theta$, so the Step (xi-f) passage from q-pilot membership in the upper ray to the ordered real inequality is explicit; the same handoff also constructs the finite extended $\Theta$-log-volume endpoint, excludes $+\infty$, identifies its real value with the hull boundary, and proves the q-pilot comparison against this real value before adjoining any $C_\Theta$-hypothesis; Lean also identifies this finite endpoint with the signed two-computation endpoint: the signed $q$-reading is the finite endpoint's q-pilot log-volume, the signed $\Theta$-reading is its real $\Theta$-value, and the formal `Corollary312Inequality` is obtained from these two shared readings;

- the Step (x) to Step (xi-g) two-computation handoff: the same data constructs the q-pilot two-computation record, with the input prime-strip reading equal to the pre-indeterminacy average and bounded by the $\Theta$-hull upper-ray value; Lean now also proves, at this route level, that the input prime-strip q-reading and the hull-side q-reading are equal, and hence that the pre-indeterminacy average is the hull-side q-reading; the Fig. 3.8 endpoint now packages both readings, their equality, both upper-ray inequalities, and hull-side q-membership in one theorem; before any $C_\Theta$ is chosen, the resulting signed two-computation endpoint already yields the formal `Corollary312Inequality`, with q- and $\Theta$-pilot magnitudes identified as the negatives of the pre-indeterminacy average and the hull boundary;

- the Step (x)-sourced $C_\Theta$ endpoint: adding $|\log(q)| > 0$ for the pre-indeterminacy q-pilot average and the displayed upper bound $-|\log(\Theta)| \leq C_\Theta|\log(q)|$ yields $C_\Theta \geq -1$; Lean also exposes the route-level chain $q_{\mathrm{avg}} \leq \Theta_{\mathrm{real}} \leq C_\Theta|q_{\mathrm{avg}}|$, hence $q_{\mathrm{avg}} \leq C_\Theta|q_{\mathrm{avg}}|$, before invoking the final ordered-real algebra; the Step (xi-f) endpoint now packages q-pilot upper-ray membership, both ordered inequalities, $C_\Theta \geq -1$, and the rejection of $C_\Theta < -1$ directly at the route level; if the q-pilot inequality is strict, Lean proves $C_\Theta > -1$; Lean now also combines this Step (xi-f) package with the Remark 3.12.2 distinct-label intertwining discipline, so the inequality endpoint is available while the q-native and $\Theta$-native labels remain unequal;

- the Step (x)-sourced statement endpoint: after adjoining the statement-level $\Theta$-subject/$q$-not-subject boundary, the same data constructs the Corollary 3.12 statement endpoint, with finite $\Theta$-branch excluding $+\infty$ and $C_\Theta \geq -1$; Lean proves that this constructed endpoint still records the $\Theta$-pilot as subject to (Ind1), (Ind2), (Ind3) and the $q$-pilot as not subject to them, and that these two status fields are unequal at the constructed endpoint; the same endpoint is now constructed directly from the $F_\ell$-cusp compatibility corridor, determinant/hull identifications, q-pilot positivity, and the displayed $C_\Theta$-bound, preserving the q-pilot average, upper-ray membership, real $\Theta$-value, and status separation; from the same $F_\ell$-sourced endpoint, Lean proves the nonnegative-$\Theta$ branch: if the (Ind3) upper bound is nonnegative, then the q-pilot average is strictly smaller than it and the endpoint has $C_\Theta \geq 0$; Lean also packages this branch as a formal 'Corollary312Inequality' value for the comparison data induced by the endpoint sign reduction; without the nonnegative-$\Theta$ assumption, the same $F_\ell$-sourced endpoint proves $q_{\mathrm{avg}} \leq C_\Theta|q_{\mathrm{avg}}|$, $C_\Theta \geq -1$, and rejection of $C_\Theta < -1$; the fixed-q two-computation version is also exposed, with the lower-bound algebra applied to $-|\log(q)|$ as determined by the input prime-strip q-computation; the signed $C_\Theta$ comparison payload is exposed from the same data, with signed $q$ equal to the pre-indeterminacy average and signed $\Theta$ equal to the hull boundary, and Lean derives the corresponding formal `Corollary312Inequality` from this signed payload while identifying the positive q- and $\Theta$-pilot magnitudes as the negatives of these two signed readings; Lean also proves that the endpoint $q$-pilot log-volume is exactly the pre-indeterminacy average and, by Step (x) invariance, also the after-(Ind1)- and after-(Ind2)-averages; thus $|\log(q)|$ is the negative of these equal averages and is positive; equivalently, the three equal averages are negative under the q-pilot positivity hypothesis; Lean also proves the displayed inequality $-|\log(q)| \leq -|\log(\Theta)|$ at this endpoint, with $q$ read as the pre-indeterminacy average and $\Theta$ read as the finite real branch; equivalently, the endpoint $q$-pilot log-volume lies in the constructed upper ray $\mathbb{R}_{\leq -|\log(\Theta)|}$; Lean also formalizes the opening reduction in the printed proof: if this finite real $\Theta$-value is nonnegative, then the strict negativity of the q-pilot log-volume gives the comparison immediately; moreover the displayed bound $-|\log(\Theta)| \leq C_\Theta|\log(q)|$ then forces $C_\Theta \geq 0$; the finite extended $\Theta$-value and the real $\Theta$-branch are identified with the Step (x) (Ind3) upper bound; the same endpoint bound holds after the (Ind1) and (Ind2) average-preserving transformations; the Step (x)-sourced endpoint also directly rejects $C_\Theta < -1$;

- base-valuation products of tensor packets, the finite $(F^\times_{\mathrm{mod}})_j$-action shadow, and a coric transfer shadow $F \to D \to D^\vdash$ preserving product log-volume without identifying Hodge-theater histories; Lean now also packages the Remark 3.12.2(i)(c) ring-structure split: additive/log-shell direct sums, multiplicative/global Frobenioid calibration, the local-shift criterion, and the guarded transfer between histories;

- the Step (xi-d)–(xi-g) shadow: determinant normalization, upper-ray membership, and equality of the two $q$-pilot log-volume computations; the determinant normalization is invariant under changing the positive tensor-power representative when the determinant log-volume is fixed; equality of the $q$-pilot log-volume with the $\Theta$-hull or determinant boundary is equivalent to supplying the corresponding reverse inequality;

- the Corollary 3.12 finiteness endpoint: the statement-level value $-|\log(\Theta)| \in \mathbb{R} \cup \{+\infty\}$ is represented by an explicit finite/$+\infty$ split, and the hull/determinant upper-ray datum forces the finite real branch, explicitly excluding the $+\infty$ branch; Lean also exposes the resulting real $\Theta$-log-volume and proves the $q$-pilot upper-ray comparison against that real value;

- the Step (xi-d) determinant normalization: the hull boundary is tied to a rank $> 1$ arithmetic-vector-bundle determinant, a positive tensor power of that determinant, and the normalized determinant log-volume; Lean now proves at the corridor endpoint that the normalized value is exactly the determinant log-volume itself, so the $\Theta$-hull log-volume is the determinant log-volume; it also records the equivalent tensor-power equations before and after normalization, and transports the q-pilot upper-ray comparison to both determinant coordinates; the same determinant/tensor-power description is now available at the finite $\Theta$-endpoint;

- the Corollary 3.12 statement boundary: the $\Theta$-pilot log-volume is typed as subject to (Ind1), (Ind2), (Ind3), while the $q$-pilot log-volume is typed as not subject to these indeterminacies;

- the extended statement endpoint: from the finite $\Theta$-branch, $0 < -|\log(q)|$, the upper-ray comparison, and the displayed hypothesis $-|\log(\Theta)| \leq C_\Theta |\log(q)|$, Lean proves $C_\Theta \geq -1$; if the finite $\Theta$-value is nonnegative, the q-pilot is strictly below it and Lean proves the sharper $C_\Theta > -1$;

- the Step (xi-f) final algebra: from $-|\log(q)| \leq -|\log(\Theta)|$, $|\log(q)| > 0$, and $-|\log(\Theta)| \leq C_\Theta |\log(q)|$, Lean proves $C_\Theta \geq -1$; from the strict inequality $-|\log(q)| < -|\log(\Theta)|$, Lean proves $C_\Theta > -1$; if $C_\Theta = -1$, Lean proves $-|\log(q)| = -|\log(\Theta)|$, hence equality with the determinant log-volume, and $-|\log(\Theta)| < 0$; hence the endpoint splits into the boundary branch $C_\Theta = -1$ or the strict branch $C_\Theta > -1$; Lean also proves the determinant-coordinate strict branch: if the fixed q-reading is strictly below the determinant boundary, then $C_\Theta > -1$; equivalently, at the determinant-coordinate endpoint, Lean separates the boundary branch where both fixed q-readings equal the determinant log-volume from the strict branch;

- the signed-payload bridge: the existing Stage 1 comparison record $q_{\text{signed}} \leq \Theta_{\text{signed}}$, together with $\Theta_{\text{signed}} \leq C_\Theta(-q_{\text{signed}})$, feeds this Step (xi-f) algebra directly;

- the Step (xi-g) endpoint bridge: the two-computation q-pilot record produces the signed comparison payload with $q_{\text{signed}}$ read from the input prime-strip log-volume and $\Theta_{\text{signed}}$ read from the hull/determinant upper-ray boundary; Lean now propagates the strict and boundary $C_\Theta$-branches to both q-computations: a strict hull-output inequality gives $C_\Theta > -1$, while $C_\Theta = -1$ forces the hull-output q-reading to equal the $\Theta$-boundary;

- the Step (xi-e)–(xi-f) fixed-value condition: in the signed two-computation endpoint, Lean proves that the input prime-strip reading and the hull-output reading are both the same real value $-|\log(q)|$, and that this value lies in the upper ray $\mathbb{R}_{\leq -|\log(\Theta)|}$; the source endpoint now also records these two fixed readings in determinant and normalized tensor-power coordinates, with equality to the determinant boundary guarded by the corresponding reverse determinant inequality;

- the fixed-value $C_\Theta$ endpoint: combining the preceding fixed upper-ray membership with $-|\log(\Theta)| \leq C_\Theta|\log(q)|$, Lean derives $-|\log(q)| \leq C_\Theta|\log(q)|$ and hence $C_\Theta \geq -1$; consequently the same hypotheses reject any $C_\Theta < -1$;

- the statement-endpoint composition: the two-computation $C_\Theta$ endpoint canonically supplies the finite $\Theta$-branch Corollary 3.12 statement endpoint once the $\Theta$-subject/$q$-not-subject indeterminacy boundary is provided;

- the opening sign reduction in the proof of Corollary 3.12: from $0 < -q_{\text{signed}}$, Lean proves that $0 \leq \Theta_{\text{signed}}$ immediately implies $q_{\text{signed}} \leq \Theta_{\text{signed}}$, matching the printed reduction to the case $-|\log(\Theta)| < 0$; the same strict-negativity argument is now available at the statement endpoint for the real $\Theta$-log-volume, and the statement endpoint now maps directly to the sign-reduction record and proves the nonnegative branch of the Corollary 3.12 comparison from it;

- the Step (xi-h) warning: for $N \geq 2$ and $\Theta < 0$, the tensor-power boundary satisfies $N\Theta < \Theta$; Lean now also proves that the original $\Theta$-boundary lies in the original upper ray but not in the tensor-power upper ray, so the replacement changes the comparison region; at the $F_\ell$-sourced corridor endpoint, this warning is tied to the finite $\Theta$-branch: the $N$-tensor-power upper ray is a strict tightening of the finite endpoint's upper ray and excludes the original real $\Theta$ value; Lean now also proves that every value in the tensor-power upper ray is strictly below the original $\Theta$-boundary; consequently, if the fixed q-reading is forced into this tensor-power upper ray while the usual $C_\Theta$-bound is assumed, Lean derives the strict branch $C_\Theta > -1$;

- the Step (xii) shadow: local Frobenioid $p_v^N$-ambiguity changes log-volume by an integer shift, while global calibration fixes exponent 0; Lean now records the exact criterion: the local shifted and unshifted log-volumes agree, and likewise the globally calibrated and local shifted values agree, exactly when the local exponent-step product is zero, equivalently when the local exponent is zero or the local prime-step log-volume is zero; for a fixed local base and nonzero local step, equality of two shifted readings is equivalent to equality of their integer exponents; at the $F_\ell$-sourced finite $\Theta$-endpoint, the globally calibrated log-volume is identified with the finite real $\Theta$-value, while any nonzero local exponent with nonzero local step gives a distinct shifted local log-volume; Lean now also couples this global calibrated value to the same $C_\Theta$-endpoint that proves $C_\Theta \geq -1$, and at that endpoint keeps the equality criterion in the decomposed exponent-or-step form; there Lean also records the ordered alternative: a positive exponent-step product places the local shifted log-volume above the calibrated value, and a negative product places it below; in the boundary branch $C_\Theta = -1$, Lean now proves that the globally calibrated value agrees with the q/determinant value while any nonzero local Frobenioid shift remains distinct from that value; conversely, identifying the shifted local value with the boundary q/determinant value is equivalent to a trivial local shift, and the sign of the local exponent-step product determines whether the shifted local value lies above or below the boundary value;

- the Remark 3.12.1(iii) arithmetic fact $\mathbb{Q}_{>0} \cap \mathbb{Z}^\times = \{1\}$, formalized as the exact characterization $0 < q \iff q = 1$ for rational integer units $q = \pm 1$;

- the Remark 3.12.1(ii) generalized-link numerical bound $C_\Theta \geq -\lambda$ for $\lambda \in \mathbb{Q}_{>0}$, with strict strengthening over the $\lambda = 1$ bound when $\lambda < 1$; Lean now proves the resulting strict conclusion $-1 < C_\Theta$ in this case, and now also splits the generalized endpoint into the boundary branch $C_\Theta = -\lambda$, where the $q^\lambda$-pilot and $\Theta$-pilot signed log-volumes are equal, or the strict branch $C_\Theta > -\lambda$;

- the $q^\lambda$-pilot endpoint algebra: from a signed comparison for $-\lambda|\log(q)|$ and the same $C_\Theta|\log(q)|$ upper bound, Lean derives $C_\Theta \geq -\lambda$, and hence $C_\Theta \geq -1$ when $\lambda \leq 1$, with the strict $-1 < C_\Theta$ conclusion when $\lambda < 1$; the $F_\ell$-sourced corridor exposes this as a conditional generalized endpoint, without asserting that the ordinary $\lambda = 1$ construction supplies all generalized links; Lean now also specializes the endpoint at $\lambda = 1$ and proves that its signed $q^\lambda$-boundary is exactly the ordinary q-pilot log-volume; the strict and boundary branch classifiers specialize to the ordinary q-pilot upper-ray comparison at $\lambda = 1$;

- the Remark 3.12.2 distinct-label implication $q$-itw $\Rightarrow$ $q$-itw $\wedge$ $\Theta$-itw/indets, with q-native and $\Theta$-native labels proved distinct; Lean now also packages this simultaneous validity together with the fixed weakened $F^{\times\mu}$-prime-strip condition, and exposes the second implication $q$-itw $\wedge$ $\Theta$-itw/indets $\Rightarrow$ $\Theta$-itw/indets as formal projection; at the $F_\ell$-sourced finite/signed endpoint, Lean combines the signed q-to-$\Theta$ comparison with this distinct-label discipline, so the comparison is available while the q-native and $\Theta$-native labels remain provably unequal; the same discipline is now attached to the Step (xi-f) $C_\Theta$-endpoint itself;

- the Remark 3.12.2 toy calculation: forgotten concrete assignments $-h$ and $-2h$ are incompatible for $h > 0$, while $-h \in \mathbb{R}_{\leq -2h+\epsilon}$ implies $h \leq \epsilon$; Lean now packages this as the full (ftoy) endpoint: membership in the upper ray, the displayed inequality $-h \leq -2h + \epsilon$, and the bound $h \leq \epsilon$; the same toy endpoint is now instantiated from the $F_\ell$-sourced finite $\Theta$-upper ray whenever the corridor q-boundary and $\Theta$-boundary are written as $-h$ and $-2h + \epsilon$;

- the Fig. 3.9 column table: Lean packages the upper-row similarities shared by the 0-column and 1-column, namely the essential $\bullet_0$ and $\circ$ roles, log-volume compatibility, and log-Kummer non-interference; the simultaneous Remark 3.12.2(iv) endpoint now records these facts for both columns, including q-pilot non-interference; Lean now also packages the q-pilot-specific combination: non-interference holds, multiradiality fails, and the log-shell treatment is tautological; it also packages the lower-row differences: $\Theta$-pilot multiradiality holds, q-pilot multiradiality does not, and the log-shell treatments differ;

- the Remark 3.12.2(v) absorption distinction: zero-column unit indeterminacy is absorbed by a hull upper ray, while one-column logarithmic conjugacy is absorbed as log-volume equality; Lean now packages these two outcomes together with the corresponding Fig. 3.9 log-shell treatments, and instantiates the zero-column hull at the finite $\Theta$-boundary supplied by the Step (x) to Step (xi) corridor; Lean also records the exact inequality-versus-equality profile: two zero-column upper bounds and one one-column equality, hence inequalities in both directions on the one-column side; on the zero-column side, Lean proves that equality with the hull boundary, after identification with the finite $\Theta$-boundary from Step (xi), requires the corresponding reverse inequality as additional data;

- the Remark 3.12.3(i) metric analogy: $\mathrm{vol}(S) > 0$ and $\chi_S = -\mathrm{vol}(S)$ imply $\chi_S < 0$, hence $\chi_S \leq 0$, together with the stored upper-semi-(Ind3) analogue;

- the Remark 3.12.4(iii) factor-$p$ analogy: theta-label normalization divides a gap log-volume by $\ell$, multiplication by $\ell$ recovers it, and the positive factor $\ell$ preserves nonpositive bounds exactly;

- the Remark 3.12.4(v) p-adic analogue: an inclusion, not isomorphism, gives $(1-p)(2g-2) \leq 0$; under the recorded hypotheses $p \geq 2$ and $g \geq 2$, Lean also proves the defect is strictly negative;

- the IUT IV, Theorem 1.10 algebraic handoff: an explicit expression for $C_\Theta$, together with $C_\Theta \geq -1$ and the remaining elementary estimates, implies the displayed $\frac{1}{6}\log(q)$ bound; Lean now exposes the exact Corollary 3.12 handoff identity $C_\Theta + 1 = A - M$, where $A$ is the arithmetic upper term and $M$ is the main logarithmic term, so $C_\Theta \geq -1$ gives $0 \leq A - M$;

- the final elementary coefficient estimates in IUT IV, Theorem 1.10: $(1-12/\ell^2)^{-1} \leq 2$ and $(1-12/\ell^2)^{-1}\frac{1}{4}(1+36d_{\mathrm{mod}}/\ell) \leq 1+80d_{\mathrm{mod}}/\ell$;

- the IUT IV, Theorem 1.10 base-change monotonicity: $\log(d_{F_{\mathrm{tpd}}}) + \log(f_{F_{\mathrm{tpd}}}) \leq \log(d_F) + \log(f_F)$ propagates through the final right-hand side;

- the IUT IV, Theorem 1.10 Step (i) identities: $n^{-1}\sum_{m=1}^{n} m = (n+1)/2$ and $n^{-1}\sum_{m=1}^{n} m^2 = (2n+1)(n+1)/6$;

- the IUT IV, Theorem 1.10 Step (ii) small-prime error: $\log(2^{11}3^35^2) \leq 21$, represented by the corresponding linear logarithmic bound;

- the IUT IV, Theorem 1.10 Step (i) $\mathrm{GL}_2$ arithmetic constants for $p = 2, 3, 5$, including the product with the quadratic factor from (E5);

- the IUT IV, Theorem 1.10 Step (iii) algebraic estimate combining the Step (ii) $K$-bound and the displayed $\log(s_Q)$-bound into $(1+4/\ell)\log(d_K, f_K) + (16/\ell)\log(s_Q) \leq (1+36d_{\mathrm{mod}}/\ell)S + 2\log(\ell) + 70$;

- the IUT IV, Theorem 1.10 Step (viii) Proposition 1.6 case split: the two cases $e^*_{\mathrm{mod}}\ell \geq \eta_{\mathrm{prm}}$ and $e^*_{\mathrm{mod}}\ell < \eta_{\mathrm{prm}}$ imply the uniform $4/3(e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}})$ bound;

- the IUT IV, Theorem 1.10 Step (viii) final absorption of $2\log(\ell) + 74$ and the prime-product term into $10(e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}})$;

- the IUT IV, Theorem 1.10 final display: the Corollary 3.12 $C_\Theta$ handoff, followed by the tripodal-to-$F$ monotonicity, gives the two displayed inequalities: first the $F_{\mathrm{tpd}}$-level $\frac{1}{6}\log(q)$ bound, then the final $F$-level bound;

- the IUT IV, Theorem A bounded-discrepancy conclusion: boundedness below of $(1+\epsilon)(\log \mathrm{diff}_X + \log \mathrm{cond}_D) - \mathrm{ht}_{\omega_X(D)}$ is packaged together with the corresponding pointwise height inequality on points of bounded degree;

- the IUT IV, Corollary 2.2(i) bounded-discrepancy chain: $\frac{1}{6}\log(q_2) \approx \frac{1}{6}\log(q_\forall) \approx \frac{1}{6}\mathrm{ht}_\infty \approx \mathrm{ht}_{\omega_X(D)}$ is packaged as three adjacent links together with the composed direct bounded-discrepancy relation;

- bounded-discrepancy transfer lemmas: pointwise upper and lower bounds move across an $\approx$ relation with the expected additive constant shift; Lean also packages simultaneous two-sided bound transfer;

- the IUT IV, Corollary 2.2(ii) condition (C1) prime-scale window: $(\log(q_\forall))^{1/2} \leq \ell \leq 10\delta(\log(q_\forall))^{1/2}\log(2\delta\log(q_\forall))$. Lean keeps the square-root and logarithmic factor explicit, proves $\log(q_\forall) \geq 0$, and derives $1 \leq 10\delta\log(2\delta\log(q_\forall))$ from the nonempty window; Lean now packages the full C1 window endpoint with these immediate consequences;

- the IUT IV, Corollary 2.2(ii) use of (C1) in Theorem 1.10: $\sqrt{h} \leq \ell$ and $80d_{\text{mod}} \leq \delta$ imply $80d_{\text{mod}}/\ell \leq \delta h^{-1/2}$; Lean packages this with the positive-denominator and nonnegative-numerator side conditions and the resulting $1 + \cdots$ coefficient inequality;

- the IUT IV, Corollary 2.2(ii) error-term use of (C1): $d^*_{\text{mod}} \leq \delta$ and $\ell \leq 10\delta h^{1/2}\log(2\delta h)$ imply $20(d^*_{\text{mod}}\ell + \eta_{\text{prm}}) \leq 200\delta^2 h^{1/2}\log(2\delta h) + 20\eta_{\text{prm}}$. Lean packages this with the nonnegativity and upper-window premises needed for the multiplication comparison;

- the IUT IV, Theorem 1.10 to Corollary 2.2(ii) first-bound composition: these two C1 replacements convert the Theorem 1.10 right-hand side into $(1 + \delta h^{-1/2})S + 200\delta^2 h^{1/2}\log(2\delta h) + 20\eta_{\text{prm}}$;

- the IUT IV, Corollary 2.2(ii) $q_2/q$ comparison: the displayed estimate $\frac{1}{6}\log(q_2) - \frac{1}{6}\log(q) \leq \frac{1}{3}h^{1/2}\log(2\delta h)$ follows through the intermediate term $h^{1/2}\log(\ell)$, with $1 \leq 6\log(\ell)$ and $3\log(\ell) \leq \log(2\delta h)$ exposed in one endpoint;

- the IUT IV, Corollary 2.2(ii) pre-$\epsilon_E$ bound for $h = \log(q_\forall)$: the Theorem 1.10 bound, the $q_2/q$ estimate, and the bounded discrepancy $B_K$ imply
$$\frac{1}{6}h \leq (1 + \delta h^{-1/2})S + (15\delta)^2 h^{1/2}\log(2\delta h) + \frac{1}{2}C_K,$$
with $C_K = 40\eta_{\text{prm}} + 2B_K$. Lean exposes the $q \to q_2 \to q_\forall$ inequalities, the absorption into $(15\delta)^2$, and the identity $\frac{1}{2}C_K = 20\eta_{\text{prm}} + B_K$;

- the IUT IV, Corollary 2.2(ii) condition (C2) inequality chain: $\frac{1}{6}\log(q) \leq \frac{1}{6}\log(q_2) \leq \frac{1}{6}\log(q_\forall) \leq (1 + \epsilon_E)(\log\text{diff}_X + \log\text{cond}_D) + C_K$;

- the IUT IV, Corollary 2.2(ii) definition of $\epsilon_E$: $\epsilon_E = (60\delta)^2 h^{-1/2}\log(2\delta h)$, represented with $h^{1/2}$ and $\log(2\delta h)$ as explicit real parameters; Lean proves positivity and $\epsilon_E \geq 5\delta h^{-1/2}$ from $\delta \geq 2$, $h^{1/2} > 0$, and $\log(2\delta h) \geq 1$, packaged as one definition endpoint;

- the IUT IV, Corollary 2.2(ii) $\epsilon_E$-error conversion: substituting the definition of $\epsilon_E$, Lean proves $(15\delta)^2 h^{1/2}\log(2\delta h) \leq \frac{1}{6}h \cdot \frac{2}{5}\epsilon_E$, packaged with $h = (h^{1/2})^2$, $h^{1/2} > 0$, and the nonnegativity premise for the logarithmic factor;

- the IUT IV, Corollary 2.2(ii) bridge to the denominator step: the pre-$\epsilon_E$ $h$-bound, the error conversion, and $\delta h^{-1/2} \leq \epsilon_E/5$ yield the preliminary inequality used to move the $\frac{2}{5}\epsilon_E\frac{1}{6}h$ term to the left; Lean packages this with log-degree nonnegativity and $1 + \delta h^{-1/2} \leq 1 + \epsilon_E/5$;

- the IUT IV, Corollary 2.2(ii) denominator step: Lean moves the term $(2\epsilon_E/5)\frac{1}{6}h$ to the left and obtains the denominator $(1 - 2\epsilon_E/5)^{-1}$, packaged with positivity of $1 - 2\epsilon_E/5$ and the preliminary inequality it acts on;

- the IUT IV, Corollary 2.2(ii) final $\epsilon_E$-absorption: from $0 < \epsilon_E \leq 1$, Lean proves
$$\frac{1 + \epsilon_E/5}{1 - 2\epsilon_E/5} \leq 1 + \epsilon_E, \qquad (1 - 2\epsilon_E/5)^{-1}C_K/2 \leq C_K,$$

using nonnegativity of $S$ and $C_K$, and hence the displayed C2 bound from the preceding denominator expression;

- the IUT IV, Corollary 2.2(ii) final $h$-bound: Lean composes the pre-$\epsilon_E$ bound, error conversion, move-left step, and absorption to prove $\frac{1}{6}h \leq (1+\epsilon_E)S + C_K$, with the square-root identity, $\epsilon_E$ definition, coefficient comparison, $0 < \epsilon_E \leq 1$, and nonnegativity hypotheses exposed in one endpoint;

- the IUT IV, Corollary 2.2(ii) final C2 bridge: using $S \leq \log \operatorname{diff}_X(x_E) + \log \operatorname{cond}_D(x_E)$, Lean derives the displayed C2 inequality chain, including the $q \to q_2 \to q_\forall$ inequalities and monotonicity under $1 + \epsilon_E \geq 0$;

- the IUT IV, Corollary 2.2(ii) comparison $\log \operatorname{diff}_X(x_E) = \log(d_{F_{\mathrm{tpd}}})$ and $\log(f_{F_{\mathrm{tpd}}}) \leq \log \operatorname{cond}_D(x_E) \leq \log(f_{F_{\mathrm{tpd}}}) + \log(2\ell)$; Lean packages these as the two corresponding sum inequalities for the tripodal and curve log sums;

- the IUT IV, Corollary 2.2 to Theorem A passage: bounded-discrepancy equivalence $\frac{1}{6}\log(q_2) \approx \mathrm{ht}_{\omega_X(D)}$, together with the C2 bound for $\frac{1}{6}\log(q_2)$, gives a lower bound for $(1+\epsilon_E)(\log \operatorname{diff}_X + \log \operatorname{cond}_D) - \mathrm{ht}_{\omega_X(D)}$; Lean packages the lower bounded-discrepancy constant, the C2 input, the induced height upper bound, and the resulting discrepancy lower bound in one endpoint;

- the IUT IV, Corollary 2.2/2.3 finite-exception passage: a lower bound on the complement of a finite exceptional set and a lower bound on the exceptional set glue to a global lower bound by taking the minimum of the two constants; Lean packages both minimum comparisons together with the resulting pointwise global lower bound;

- the IUT IV, Corollary 2.3 Diophantine inequality: Lean records the GenEll transfer as an explicit hypothesis and derives the global bounded-discrepancy lower bound for $(1+\epsilon)(\log \operatorname{diff}_X + \log \operatorname{cond}_D) - \mathrm{ht}_{\omega_X(D)}$, packaged with $d > 0$, $\epsilon > 0$, the hyperbolic-curve premise, and the compact-subset Theorem A premise;

- comparison levels: pointwise representative, aggregate representative, balanced sign-compatible, and hull/log-volume; Lean proves that the final weighted-theta route is hull/log-volume level and rejects the balanced sign-compatible level, including the balanced negation transport;

- endpoint audits for the 3.11.5 $\Rightarrow$ 3.12 comparison chart;

- nonarchimedean and archimedean (Ind3) local orientations;

- ordered real-line alignment with cancellation only after scale equality;

- a canonical unit-scale ordered alignment from a nonarchimedean (Ind3) entry;

- the local-object Hodge-descent square-weight boundary: transported zero-column and cusp-class local objects have the same finite log-volume as the packet object, while square/full-label preservation obligations remain explicit;

- the square/full-label Hodge/SHE transport audit: a pointwise upper bound on the transported target full-label log-volumes transfers to the source square-weighted average once full-label log-volume, square weight, and total-weight preservation are supplied; the same transfer is now available inside the structured-SHE wrapper, and the resulting upper bound is exposed for the square-weighted average in the $F_\ell$ cusp-label route after matching the source profile and source log-volume; consequently, structured-SHE target full-label bounds by the $\Theta$-source average feed the cusp-class container route all the way to $q_{\mathrm{signed}} \leq \Theta_{\mathrm{signed}}$; the same final comparison is now available directly from the primitive factored square/full-label SHE obligations, without requiring a prebuilt transport-audit argument;

- a Gaussian-to-factored-SHE bridge: source and target degree-level Gaussian evaluations with equal environment degree supply the full-label value-preservation branch of the factored square/full-label SHE obligations; with separate coordinate-square and full-label-map preservation, Lean exposes the resulting pointwise comparison level, transported-average equality, forced identity of the representative square coordinate map, and retained separation of the two Hodge histories; the experiment route then reaches $q_{\mathrm{signed}} \leq \Theta_{\mathrm{signed}}$ directly from Gaussian target bounds; since the absolute-label exponent is nonnegative, Lean also derives those target bounds in the branch where the target environment degree is nonpositive and the $\Theta$-source average is nonnegative; conversely, the all-label Gaussian target-bound formulation forces the $\Theta$-source average to be nonnegative, since the core label has Gaussian degree 0; after removing the core label, Lean proves that every nonzero Gaussian target value is bounded by the $\Theta$-source average if the target environment degree is nonpositive and is itself bounded by that average; Lean also forms the corresponding finite average over the nonzero $ZMod$ carrier, proves the same upper bound for this nonzero average, and proves the exact rescaling between this auxiliary nonzero average and the uniform $F_\ell$-average with zero Gaussian term; because the square-weighted route assigns weight 0 to the core label, Lean now feeds nonzero-only target bounds through the square/full-label transport audit all the way to $q_{\mathrm{signed}} \leq \Theta_{\mathrm{signed}}$; this nonzero Gaussian environment-comparison route is now a source-layer theorem, not only an experiment wrapper;
- a bridge from factored square/full-label SHE obligations to the Hodge-descent (Ind3) route.

## 5 Current Lean Results

The current first-pass experiment report evaluates to:

$$\begin{aligned}
&\text{route theorem flags} && = \texttt{true}, \\
&\text{collapse-obstruction flags} && = \texttt{true}, \\
&\text{Gaussian/procession finite-model flags} && = \texttt{true}, \\
&\texttt{balancedLevelRejectedAtFinalRouteTheoremAvailable} && = \texttt{true}, \\
&\texttt{disputeSettledByCurrentStage} && = \texttt{false}.
\end{aligned}$$

Here the route flags are ordered real-line, nonarchimedean entry, and factored SHE availability. The obstruction flags are raw-cancellation failure, label-independent $j^2$ failure, and sign-quotient representative descent failure. The Gaussian/procession flags include finite tensor-packet, monoid action, normalized splitting-generator average, the explicit square-sum formula for the generator total and normalized delta, the resulting conditional monotonicity of the acted tensor packet, canonical positive square-weight profile, the constant-average check, base-valuation product, coric-transfer, determinant-normalization, upper-ray, $q$-pilot two-computation, the Step (x) averaged-log-volume indeterminacy corridor, the refined Step (x) bounds for the before-, after-(Ind1)-, and after-(Ind2)-averages, tensor-power warning, and global Frobenioid-calibration shadows, plus the finiteness endpoint for $-|\log(\Theta)| \in \mathbb{R} \cup \{+\infty\}$, including extraction of the finite real $\Theta$-value, the direct Corollary 3.12 statement endpoint with $\Theta$-subject/$q$-not-subject pilot bookkeeping, the Step (x)-sourced upper-ray, q-pilot two-computation, statement endpoint, propagation of the same pilot-status boundary to that endpoint, proof that the endpoint $\Theta$- and $q$-statuses are distinct, identification of its $q$-pilot real with the Step (x) pre-, after-(Ind1)-, and after-(Ind2)-averages, positivity of $|\log(q)|$ at that endpoint, and direct $-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}$, $-|\log(q)| \leq -|\log(\Theta)|$, finite-$\Theta$ exclusion of $+\infty$, identification of the finite extended $\Theta$-value with the Step (x) (Ind3) upper bound, the nonnegative-$\Theta$ opening reduction including the stronger $C_\Theta \geq 0$ branch, and $C_\Theta \not< -1$ conclusions, the Step (xi-g)-to-(xi-f) endpoint bridge giving $C_\Theta \geq -1$ from the two-computation q-pilot record, the Step (xi-e)–(xi-f) fixed $-|\log(q)|$ membership check and its direct $C_\Theta \geq -1$ consequence, the composition into the statement-level endpoint, and positive-rational unit rigidity. The generalized-link flag records the rational parameter inequality $C_\Theta \geq -\lambda$, now with the corresponding $q^\lambda$-pilot real-valued derivation. The distinct-label flag records the logical intertwining display in Remark 3.12.2, keeps the weakened abstract prime-strip condition fixed, and rejects collapse of the q-native and $\Theta$-native labels; the final Step (xi-f) inequality package is now checked under this same label separation. The toy-model flag records the resulting (ftoy) upper-ray endpoint and arithmetic bound $h \leq \epsilon$. The column-table flag records the Fig. 3.9 theta/q similarity-difference split: upper similarities are common to both columns, while multiradiality and log-shell treatment distinguish the 0-column from the 1-column; the upper similarities are now exposed as a simultaneous zero/one-column theorem. The absorption flag records the zero-column upper-bound versus one-column equality distinction, tied to the two Fig. 3.9 log-shell treatments; the one-column equality is now exposed together with its two induced inequalities, while zero-column hull equality is equivalent to adding the missing reverse bound; the same guard is now stated after replacing the hull by the finite $\Theta$-boundary. The metric flag records the Gauss–Bonnet sign calculation used as analogy for the upper semi-compatibility inequality, including the nonpositive inequality form following from strict negativity. The log-link-juggling flag records the Remark 3.12.2(iii) source of (Ind3): unchanged procession-normalized averages through (Ind1) and (Ind2), plus a one-sided upper-semi bound and matching place families. The factor-$p$ flag records the $F_\ell$-label averaging denominator and the order-preservation of this positive normalization. The Frobenius-derivative flag records the degree inequality from inclusion and the sharper strict negativity of the degree defect under $p \geq 2$, $g \geq 2$. The IUT IV flag records the downstream Theorem 1.10 use of the Corollary 3.12 lower bound on $C_\Theta$, including the identity $C_\Theta + 1 = A - M$, the resulting nonnegative gap $0 \leq A - M$, the final coefficient estimates, and the tripodal-to-$F$ log-degree monotonicity. The Step (i) flag records the two elementary sum identities used in the leading-term computation. The Step (ii) small-prime flag records the additive error bound 21. The $\mathrm{GL}_2$ flag records the finite cardinality constants feeding that bound. The Step (iii) flag records the $\log(s_Q)$ coefficient absorption. The Step (viii) flag records the prime-product case split used before the final upper bound for $C_\Theta$, plus the final coarse error absorption. The final display flag records the composed Theorem 1.10 inequality with $F$-level log-degree terms. The Theorem A flag records the bounded-discrepancy height/log-different/log-conductor inequality. The Corollary 2.2 flag records composition of the displayed bounded-discrepancy chain. The transfer flag records movement of pointwise bounds across bounded-discrepancy equivalence. The Corollary 2.2(C1) flag records the displayed prime-scale window and its elementary consequences. The Corollary 2.2(C2) flag records the displayed three-step logarithmic inequality chain. The C1 coefficient flag records the replacement of $80d_{\mathrm{mod}}/\ell$ by $\delta h^{-1/2}$ in the Theorem 1.10-to-C2 proof. The C1 error-term flag records the corresponding replacement of $20d^*_{\mathrm{mod}}\ell$ by $200\delta^2 h^{1/2}\log(2\delta h)$. The first-bound flag records the composition of those replacements with Theorem 1.10. The $\epsilon_E$-definition flag records the source formula and lower estimate for $\epsilon_E$. The $\epsilon_E$-move-left flag records introduction of the denominator $(1 - 2\epsilon_E/5)^{-1}$. The $\epsilon_E$-absorption flag records the final denominator removal in C2. The $\epsilon_E$-error flag records the conversion of the

$(15\delta)^2$-term to the $\frac{2}{5}\epsilon_E\frac{1}{6}h$ term. The final $h$-bound flag records the full composition through absorption. The final C2 bridge flag records the last monotonic replacement of the tripodal log-degree sum by the curve different/conductor sum. The $h$-to-move-left flag records the bridge from the pre-$\epsilon_E$ bound to the denominator-stage preliminary inequality. q2/q flag records the displayed comparison of $\log(q_2)$ and $\log(q)$. The pre-$\epsilon_E$ $h$-bound flag records the subsequent absorption of the bounded $q_\forall/q_2$ discrepancy into $C_K$. The log-different/log-conductor flag records the bridge from tripodal field terms to curve functions. The Corollary 2.2-to-Theorem A flag records the formal bounded-discrepancy transfer from C2 to the Theorem A inequality. The finite-exception flags record the lower-bound gluing used when finite exceptional sets are enlarged in Corollary 2.2 and then passed toward Corollary 2.3. The Corollary 2.3 flag records the final GenEll-transfer interface for the Diophantine inequality. The theorem-level route is:

$$\begin{array}{ccccc}
\text{nonarch. (Ind3) entry} & \to & \text{canonical ordered } \mathbb{R}\text{-alignment} & \to & \text{source/target alignment} \\
 & & \downarrow & & \downarrow \\
\text{factored SHE transport} & \to & \text{Hodge-descent audit} & \to & q_{\text{signed}} \leq \Theta_{\text{signed}}.
\end{array}$$

The route remains conditional on the displayed hypotheses. The local-object Hodge-descent layer now exposes equality of the transported finite log-volumes before square/full-label preservation is supplied. The square/full-label audit now also transports target pointwise upper bounds back to the source square-weighted average, including through the structured-SHE route wrapper, and the experiment layer exposes the resulting final q-to-$\Theta$ comparison theorem. The same experiment-level theorem is now exposed for primitive factored square/full-label SHE obligations and for degree-level Gaussian evaluations after the explicit preservation branches are supplied. In the nonpositive-environment/nonnegative-$\Theta$-average branch, the Gaussian pointwise target bound is derived from the sign of the $q^{j^2}$-exponent rather than assumed. Lean also proves that this all-label Gaussian bound cannot hold when the $\Theta$-source average is negative; under nonpositive environment it is equivalent to nonnegativity of that average. Lean also packages the resulting separation: for negative $\Theta$-source average, the all-label bound fails, while the nonzero-label bound still holds whenever the Gaussian environment degree is bounded by that same average. With the remaining endpoint hypotheses, this same negative $\Theta$-branch still yields the final $q_{\text{signed}} \leq \Theta_{\text{signed}}$ route through the nonzero-bound endpoint; the target nonpositive-environment hypothesis is then available, but the nonnegative-$\Theta$ hypothesis of the nonpositive/nonnegative all-label endpoint is impossible. The source-supplied nonzero/environment formulation gives the same conclusion and rejects both source and target all-label bounds. More generally, Lean decomposes the all-label Gaussian bound exactly into nonnegativity of the upper bound at the zero label and the corresponding nonzero-label bound. The corresponding identity-coordinate final route can now be stated directly from canonical $j = 1$ equality, nonpositive target environment, and nonnegative $\Theta$-source average; if the all-label Gaussian target bound is supplied directly, the same endpoint no longer needs the nonpositive/nonnegative branch hypotheses, and its proof also uses only the nonzero-label part of the supplied all-label target bound. The nonzero environment route also has a source-side variant: canonical $j = 1$ equality transports nonpositivity and the comparison with the $\Theta$-source average from the source environment degree to the target environment degree. Lean also proves that the nonzero target-bound predicate is then equivalent to this source-environment comparison, hence to the source nonzero-bound predicate under source nonpositivity, and that the all-label predicate is equivalent to this source comparison plus nonnegativity at the zero label. The final all-label q-to-$\Theta$ endpoint now consumes this source-side package directly through the source-environment comparison; the zero-label nonnegativity component is retained in the API but not used by this final proof. It can alternatively derive that package from a source all-label Gaussian bound under source nonpositivity; the proof now uses only the nonzero-label part of this source all-label input. The source all-label predicate is itself equivalent, under source nonpositivity, to the conjunction of $0 \le \Theta_{\mathrm{avg}}$ and the source-environment upper bound. Hence, under canonical $j = 1$ preservation and the two nonpositive branches, the source and target all-label Gaussian predicates are equivalent. It can also consume the decomposed zero/nonzero input directly; this endpoint now ignores the zero-label nonnegativity component and proceeds through the nonzero factored route. In the negative $\Theta$-branch, Lean proves that this zero-label nonnegativity component is impossible, while the source environment/nonzero route still gives the final comparison. For nonzero labels, Lean proves the complementary bound: the absolute exponent is at least 1, so a nonpositive environment degree below the $\Theta$-source average bounds every nonzero Gaussian component and hence their finite nonzero-label average. The final nonzero route may now consume the source nonzero Gaussian predicate directly after canonical $j = 1$ preservation and source nonpositivity; Lean reduces this input to the source-environment upper bound before invoking the source-environment endpoint. The square-weighted transport route has also been strengthened from an all-label pointwise bound to a weighted-summand bound, so the zero label requires no target upper bound. The experiment theorem now delegates to the source-layer q-to-$\Theta$ theorem for this nonzero Gaussian environment-comparison route. Separately, Lean proves that the uniform $F_\ell$-average of the Gaussian shadow is $(\ell - 1)/\ell$ times the auxiliary nonzero-carrier average, fixing the denominator at $\ell$ when the zero label is present.

# 6 Audit Findings

No formal statement currently proves IUT III, Corollary 3.12 from Mochizuki's published hypotheses. The strongest positive result is conditional: if the hull/log-volume upper-ray comparison supplies $-|\log(q)| \le -|\log(\Theta)|$, then Lean checks the formal Step (xi-f) conclusion $C_\Theta \ge -1$; if the upper-ray membership is strict, Lean checks the sharper $C_\Theta > -1$, while the boundary value $C_\Theta = -1$ collapses the q-pilot and $\Theta$-boundary log-volumes to equality and forces the $\Theta$-boundary to be negative. If the nonarchimedean (Ind3) entry, ordered real-line alignment, and Hodge/SHE square-weight transport audit are supplied, Lean checks the final signed $q$-to-$\Theta$ inequality feeding that algebra. If the corresponding primitive factored square/full-label SHE obligations are supplied instead, Lean first constructs the transport audit and checks the same final inequality. If the input is a pair of degree-level Gaussian evaluations with equal environment degree, Lean constructs the full-label log-volume compatibility branch before invoking the factored route. The branch with nonnegative $\Theta$-source average is formally easier: nonnegativity of $j^2$ makes every nonpositive environment-degree Gaussian target value bounded above by that average. The all-label version cannot be the negative branch: the core label contributes 0, so a negative upper bound is impossible; Lean now packages this as an exact equivalence with nonnegativity of the upper bound and as a separation theorem from the nonzero-label bound under the environment comparison; the separated nonzero-label branch also feeds the final $q$-to-$\Theta$ theorem, both from target-side and source-side environment data. In the source-side version, Lean rejects both all-label predicates while retaining the final comparison. For the target-side nonpositive/nonnegative endpoint, Lean isolates the failed premise as $0 \le \Theta_{\mathrm{avg}}$, not target nonpositivity. On the nonzero-label part, the formal obstruction disappears: exponent $\ge 1$ gives the expected bound from the environment comparison. The square-weighted final route now accepts this nonzero-only bound, since the core summand has weight 0. This gives a conditional source-level q-to-$\Theta$ route from degree-level Gaussian data, coordinate-square preservation, full-label-map preservation, equal source/target environment degree, and environment degree bounded above by the $\Theta$-source average. In the identity-coordinate case, Lean now supplies the coordinate-square and full-label-map preservation branches internally, so this route only asks for equal Gaussian environment degree and the nonpositive environment comparison; equivalently, the equal-environment input may be replaced by equality of the canonical $j = 1$ Gaussian components, and the target-side environment comparison may be supplied on the source side after the same equality; the nonzero target-bound predicate is equivalent to this source-side comparison, and the all-label target predicate is equivalent to this comparison together with $0 \le \Theta_{\mathrm{avg}}$. The corresponding final theorem now uses these source-side inputs directly rather than asking for the all-label target predicate separately; a further variant derives them from a source all-label Gaussian bound, and Lean classifies that source all-label predicate as exactly $0 \le \Theta_{\mathrm{avg}}$ plus the source-environment bound; the target and source all-label predicates are consequently equivalent under the same canonical $j = 1$ and nonpositivity hypotheses. Lean also proves that, in the nonpositive environment branch, the nonzero coordinate target-bound formulation is equivalent to this environment bound against the $\Theta$-source average and, with source nonpositivity, to the source nonzero predicate; the necessity direction does not use the nonpositive hypothesis, since the canonical $j = 1$ component is the environment degree itself. The final identity-coordinate route may now be stated directly from canonical $j = 1$ equality and the nonzero Gaussian target-bound predicate; after source nonpositivity, the corresponding source nonzero-bound predicate supplies the same endpoint through the source-environment classification. In the all-label branch, the analogous direct endpoint may either use canonical $j = 1$ equality plus the all-label Gaussian target-bound predicate, or derive that predicate from nonpositive target environment and nonnegative $\Theta$-source average. Lean also records the inclusion from the all-label predicate to the nonzero-label predicate used by the weighted-summand route, and the converse after adding the zero-label nonnegativity condition; the corresponding direct endpoint is exposed with this decomposed input. The target all-label endpoint is now implemented by this same nonzero/square-weighted route rather than by a full all-label factored bound. The zero/nonzero average split is now exact: the nonzero carrier has cardinality $\ell - 1 = 2j_{\max}$, the signed carrier is a twofold cover of the nonzero absolute half-range, the zero Gaussian degree is 0, and the all-$F_\ell$ average is $(\ell-1)/\ell$ times the nonzero absolute average; consequently the $F_\ell$-coordinate and full $|F_\ell|$-absolute averages differ strictly for nonzero environment degree. Lean also now proves the local distinct-label fact emphasized in IUT II: the $j = 1$ and $j = 2$ theta/Gaussian components coincide if and only if the environment degree vanishes. The remaining task is to replace the current degree-level Gaussian shadow by the actual Hodge–Arakelov Gaussian/Frobenioid construction behind this environment comparison.

The strongest negative result is also limited: in the current $\mathbb{F}_\ell$-label model, Lean rejects a single label-independent scale accounting for all representative $j^2$ factors, and it rejects balanced sign-compatible evidence as the final hull/log-volume comparison level. Lean also rejects direct descent of arbitrary raw representative $j^2$ theta-degree data to $|F_\ell|$, while the underlying $F_\ell$-label model, additive torsor laws, sign-unit quotient, and canonical cusp-class bridge are now packaged as source-side endpoints, accepting the half-range exponent convention stated in IUT II. Lean also checks that an $N$-th tensor-power boundary is a strict strengthening for $N \ge 2$ and negative pilot log-volume. It also checks the Step (xii) numerical warning: nonzero local Frobenioid shifts change log-volume, whereas global calibration restores the unshifted value. The positive rational unit rigidity fact in Remark 3.12.1(iii) is formalized directly over $\mathbb{Q}$. For generalized links, Lean records that $\lambda < 1$ makes $-\lambda > -1$. At $\lambda = 1$, the generalized endpoint reduces to the ordinary q-pilot boundary. Lean also checks that the Remark 3.12.2 simultaneous-intertwining step depends on distinct labels rather than an unlabeled identification, while the weakened prime-strip condition is retained. In the accompanying toy model, Lean proves the final upper bound exactly from the upper-ray membership hypothesis. Lean also records that the q-pilot column lacks the theta-side multiradiality used in Theorem 3.11(iii)(c). For Remark 3.12.2(iv), Lean records that the essential $\bullet_0$ and $\circ$ roles, log-volume compatibility, and log-Kummer non-interference hold on both the 0-column and the 1-column. For Remark 3.12.2(iii), Lean packages the log-link juggling as the source of the upper-semi

(Ind3) endpoint: the averaged value is fixed through (Ind1) and (Ind2), then bounded from above. For Remark 3.12.2(v), Lean separates hull absorption of unit shifts from log-volume equality after logarithmic-conjugate compatibility, and records that q-pilot non-interference does not supply q-pilot multiradiality. It also checks that zero-column hull upper bounds do not become hull equalities without reverse inequalities, even after the hull is identified with the finite $\Theta$-boundary in Step (xi). For Remark 3.12.3(i), Lean proves the stated negative Euler-characteristic sign from positivity of metric volume. For Remark 3.12.4(iii), Lean checks the $\ell$-denominator normalization identity for the theta-label averaging analogy. For Remark 3.12.4(v), Lean proves the displayed nonpositive degree defect from $p \geq 2$ and $g \geq 2$. For IUT IV, Theorem 1.10, Lean proves the algebraic implication from the computed $C_\Theta$ expression and $C_\Theta \geq -1$ to the final $\frac{1}{6}\log(q)$ estimate, conditional on the separate elementary estimates. Lean now also proves the two displayed final elementary coefficient estimates from $\ell \geq 5$. It also proves the final monotonic replacement of the tripodal log-degree sum by the larger $F$-level log-degree sum, conditional on the two component monotonicity inequalities cited from Proposition 1.3(i). Lean also proves the Step (i) sum and square-sum identities over $\mathbb{R}$. For Step (ii), Lean proves the small-prime additive error bound from the displayed logarithmic inequalities. Lean also checks the Step (i) $\mathrm{GL}_2$ constants for $p = 2, 3, 5$ and their product with the quadratic extension factor. For Step (iii), Lean proves the displayed coefficient absorption from the source hypotheses $d_{\text{mod}} \geq 1$, $\ell \geq 5$, $2\log(\ell) \leq \ell$, and nonnegativity of the tripodal log-degree sum. For Step (viii), Lean checks that the two Proposition 1.6 cases imply the uniform prime-product bound used in the final $C_\Theta$ estimate. Lean also checks the subsequent absorption into the coefficient 10 of $e^*_{\text{mod}}\ell + \eta_{\text{prm}}$. Combining the Corollary 3.12 handoff with the tripodal-to-$F$ monotonicity, Lean proves the two-line final display of Theorem 1.10 at this real-valued shadow level: the intermediate $F_{\text{tpd}}$-level bound and the final $F$-level bound. For Theorem A, Lean records the exact bounded-discrepancy conclusion and packages it with the equivalent pointwise height bound from the lower-bound hypothesis. For Corollary 2.2(i), Lean defines equality up to bounded discrepancy, proves reflexivity, symmetry, transitivity, nonnegative scaling, and composes the displayed chain from $\log(q_2)$ to $\mathrm{ht}_{\omega_X(D)}$; the chain endpoint records both the adjacent links and the composite relation. Lean also proves that upper and lower pointwise bounds transfer across such a bounded-discrepancy equivalence with the appropriate constant shift, including a combined two-sided transfer theorem. For Corollary 2.2(ii)(C1), Lean records the displayed window for $\ell$ and derives nonnegativity of $\log(q_\forall)$, positivity of $(\log(q_\forall))^{1/2}$, and $1 \leq 10\delta\log(2\delta\log(q_\forall))$, packaged as a single C1 endpoint. Lean also verifies the C1 coefficient replacement $80d_{\text{mod}}/\ell \leq \delta h^{-1/2}$ used immediately after applying Theorem 1.10, together with the positivity/nonnegativity facts needed for the division comparison. Lean also verifies the C1 upper-window replacement for the Theorem 1.10 error term $20(d^*_{\text{mod}}\ell + \eta_{\text{prm}})$, including the nonnegativity and upper-window facts used in the product estimate. Combining these, Lean proves the first displayed post-Theorem-1.10 bound in the proof of Corollary 2.2(ii). Lean also proves the displayed comparison $\frac{1}{6}\log(q_2) - \frac{1}{6}\log(q) \leq \frac{1}{3}h^{1/2}\log(2\delta h)$ from the intermediate $h^{1/2}\log(\ell)$ bounds and their two logarithmic side inequalities. Lean then combines this with the bounded $q_\forall/q_2$ discrepancy to prove the displayed pre-$\epsilon_E$ bound for $\frac{1}{6}h$, with the $q \to q_2 \to q_\forall$ transfers, $(15\delta)^2$ absorption, and $\frac{1}{2}C_K$ rewrite exposed as named subclaims. For Corollary 2.2(ii), Lean proves the final $\frac{1}{6}\log(q)$ bound from the displayed $\log(q)$, $\log(q_2)$, and $\log(q_\forall)$ chain. Lean records the proof's definition of $\epsilon_E$ and verifies the stated lower estimate $\epsilon_E \geq 5\delta h^{-1/2}$, including the positivity facts used in the inverse-square-root comparison. Lean also verifies the source conversion $(15\delta)^2 h^{1/2}\log(2\delta h) \leq \frac{1}{6}h \cdot \frac{2}{5}\epsilon_E$, with $h = (h^{1/2})^2$ and the required nonnegativity facts explicit. Combining this with $\delta h^{-1/2} \leq \epsilon_E/5$, Lean proves the preliminary inequality needed for the move-left step, including the nonnegative log-degree premise used to scale the coefficient comparison. Lean then verifies the algebraic move-left step that introduces the denominator $(1 - 2\epsilon_E/5)^{-1}$, exposing denominator positivity and the input preliminary inequality. Lean also proves the final $\epsilon_E$-absorption step used just before C2: the intermediate expression with denominator $1 - 2\epsilon_E/5$ is bounded by $(1 + \epsilon_E)$ times the log-degree sum plus $C_K$, with denominator $\geq 1/2$ and nonnegativity of $S$ and $C_K$ explicit. Lean now composes these steps into the final displayed $h = \log(q_\forall)$ bound, with all square-root, $\epsilon_E$, coefficient, and nonnegativity premises exposed in one endpoint. Lean also derives the displayed C2 chain from this $h$-bound and the tripodal-to-curve log-degree comparison, recording the $q \to q_2 \to q_\forall$ inequalities and the monotonicity condition $1 + \epsilon_E \geq 0$. It also proves the two sum inequalities relating the tripodal different and conductor terms to $\log \operatorname{diff}_X + \log \operatorname{cond}_D$, packaged with the defining equality and conductor inequalities. For the Corollary 2.2 to Theorem A passage, Lean combines the bounded-discrepancy lower estimate for $\frac{1}{6}\log(q_2) - \operatorname{ht}_{\omega_X(D)}$ with the C2 upper bound for $\frac{1}{6}\log(q_2)$, deriving both the corresponding upper bound for the height and the lower bound for $(1 + \epsilon_E)(\log \operatorname{diff}_X + \log \operatorname{cond}_D) - \operatorname{ht}_{\omega_X(D)}$. For the Corollary 2.2/2.3 exceptional-set passage, Lean proves that a lower bound off a finite exceptional set and a lower bound on the exceptional set imply a global lower bound, and specializes this to the Theorem A discrepancy. The endpoint records both minimum comparisons and the resulting pointwise bound. For Corollary 2.3, Lean proves the stated bounded-discrepancy conclusion from an explicit GenEll transfer hypothesis; GenEll itself is not formalized here. The endpoint exposes $d > 0$, $\epsilon > 0$, hyperbolicity, and the compact-subset Theorem A premise. Finally, Lean combines the Corollary 2.2(i) bounded-discrepancy height comparison with the C2 bound to construct the Theorem A lower-bound shadow.

# 7 Missing Mathematics

The present code does not yet formalize:

- initial $\Theta$-data as in IUT I;
- actual Hodge theaters, Frobenioids, prime strips, log-shells, or $\Theta$-links;
- the Hodge–Arakelov theta evaluation and Gaussian monoids of IUT II;
- the construction of the actual theta values, Gaussian/splitting monoids, $F^{\times}_{\mathrm{mod}}$-actions, Kummer isomorphisms, mono-analyticization, and Frobenioids behind the present shadows;
- the actual construction of log-shells and tensor packets attached to mono-analytic prime-strips in IUT III, Theorem 3.11;
- the genuine holomorphic hull and determinant operations of Step (xi), beyond the present finite determinant/upper-ray/two-computation/tensor-power shadows;
- the global realified Frobenioids of Step (xii), beyond the present integer shift and calibrated-$C_\Theta$ shadow;
- the arithmetic-divisor and analytic local log-volume construction behind the IUT IV Theorem 1.10 numerical estimates, beyond the present source-side interface records;
- the construction of the normalized height, log-different, and log-conductor functions used in IUT IV, Theorem A;
- the construction of the bounded-discrepancy equivalences asserted in IUT IV, Corollary 2.2(i);
- a proof that the current Hodge/SHE transport obligations follow from Mochizuki's full hypotheses.

## 8 Next Work

The finite $q^{j^2}$, splitting-generator, and Gaussian-to-factored-SHE packaging endpoints are now in place. The next significant formal step is to replace current explicit Hodge/SHE transport assumptions by constructions closer to the papers:

$$\begin{gathered}\text{theta values } q^{j^2} \quad \checkmark \text{ finite square-weight endpoint}\\ \rightsquigarrow \ \text{Gaussian and splitting monoids} \quad \checkmark \text{ generator action endpoint}\\ \rightsquigarrow \ \text{factored square/full-label preservation} \quad \checkmark \text{ Gaussian endpoint}\\ \rightsquigarrow \ \text{hull/log-volume route.}\end{gathered}$$

The next step is to construct the Hodge/SHE preservation hypotheses themselves.

## References


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